OAKLAND COMMUNITY COLLEGE®

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2140
FactoryTalk View Machine Edition HMI LABS

John Sefcovic
Oakland Community College

TABLE OF CONTENTS

HUMAN MACHINE INTERFACE (HMI) FactoryTalk View ME HMI LABS

LAB: MOVE INSTRUCTION

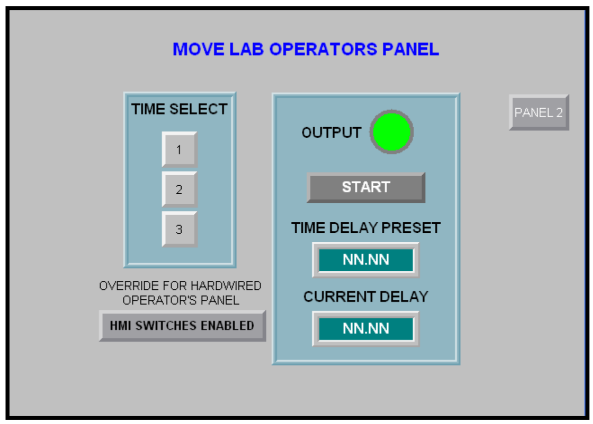
Modify the HMI template TEMPLATE MOVE_HMI for the MOVE INSTRUCTION, PART 2 lab.

See desktop PDF document <u>FactoryTalk View Studio Machine Edition Procedures</u>.

Use Application Manager to make a copy with a new name MOVE_HMI_*last name*.

Configure communication with Studio 5000 offline-file/processor with a shortcut named **HMI_DATA**.

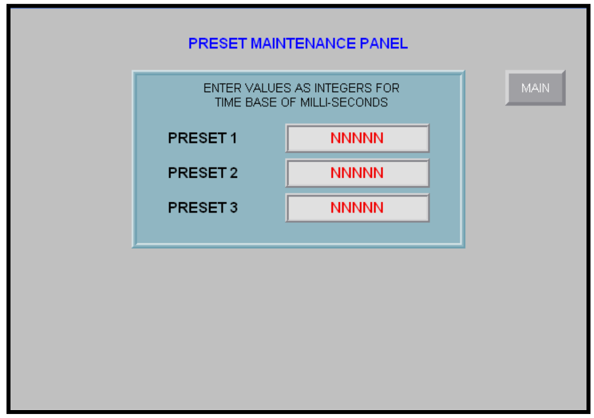
OPERATORS PANEL 1



- Delete the unused objects.
- Edit to change the static text captions.
- Edit to change button objects captions.
- Add button values to TIME SELECT buttons.
- Add connection tags to (not text) the pushbutton and indicator objects. See page 4.
- Add connection tags for the numeric display objects.
- Add the Maintained pushbutton, Normally Open, and the Text over the button as shown.
    - The caption must read “HMI SWITCHES ENABLED” with black and bold text when the button’s value is 1.
    - The caption must read “HMI SWITCHES DISABLED” with red and bold text when the button’s value is 0.
    - See page 4 for the HMI ENABLED buttons back colors.
- Edit the goto navigation for operator’s panel 2.
- Format screen appearance to match screen shown.

OPERATORS PANEL 2

- Delete the unused objects.
- Edit to change the text objects and object's captions.
- Add connection tags to (not text) the numeric input objects. See page 3.
- Edit the goto screen navigation for display operator's panel 2.
- Format screen appearance to match screen shown.
-

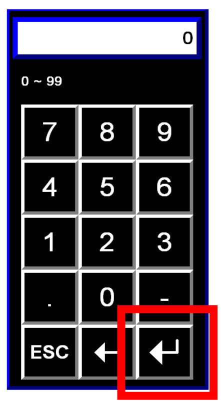


Notice the presets are entered in milliseconds.

The "NNN" are number values displayed in development, values will appear in runtime.

When selecting Numeric Input during runtime a keypad will appear.

Use cursor to select numbers.



ENTER key

---

continue

PLC TAGS ASSIGNED TO HMI OBJECTS

PANEL 1 TAGS

Tag	Object	Tag	Object
HMI_PB	Time Select Buttons	OUTPUT	Output Indicator
HMI_ENABLE_DISABLE	Disable Button	START	Start Pushbutton
		HMI_PRE	Panel Timer Preset
		HMI_ACC	Panel Timer Accumulated

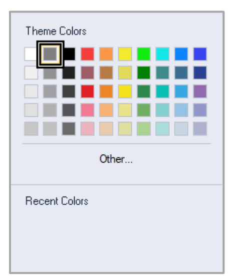
PANEL 2 TAGS

Tag	Object
PRE_SET_1	Preset No.1
PRE_SET_2	Preset No.2
PRE_SET_3	Preset No.3

HMI SWITCH ENABLED BUTTON - BACK COLORS

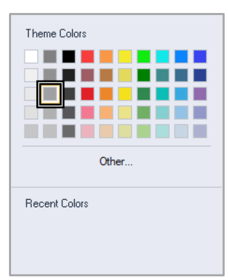
State 0:

Change Back Color to a Dark Gray.



State 1:

Change Back Color to a Light Gray.



LAB: TIMED PROCESS OUTPUT

Modify the HMI template TEMPLATE PROCESS OUTPUT for the TIMED PROCESS OUTPUT lab.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures.

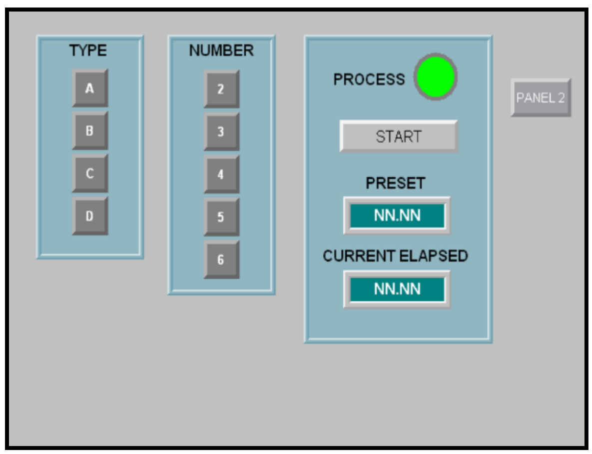
Use Application Manager to make a copy with a new name PROCESS_OUTPUT_*last name*.

Configure communication with Studio 5000 off line-file/processor with a shortcut named **HMI_DATA**.

PLC TAG CONNECTIONS

The PLC Tag connection for all (except the START tag) objects will use the tag name starting with **HMI_**.

OPERATORS PANEL 1



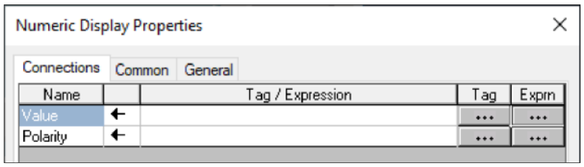
- Delete the unused objects.
- Edit to change the static text captions.
- Edit to change button objects captions.
- Add button values to sheet TYPE.
- Add button values to NUMBER of sheets.
- Add connection tags to (not text) the pushbutton and indicator objects.
- Add connection tags for numeric display objects.
- Enter the numeric displayed objects expression to convert milliseconds time actual time. See page 5.
- Edit the go to navigation for the display operator's panel 2.
- Format screen appearance to match screen shown.

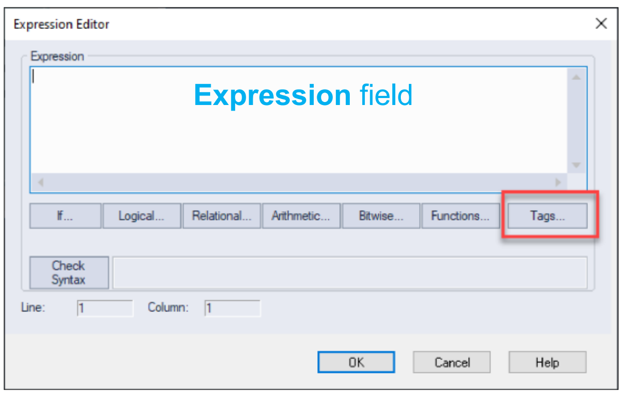
HMI PRESET and CURRENT ELAPSED NUMERIC DISPLAY OBJECTS

The **Alias Tag** for the Process Timer's preset and accumulated values is divided by 1000 when shown in the corresponding Numeric Display object.

***alias tag for operator preset or accumulated*/1000**

In the Connections dialog, select **Exprn** button.

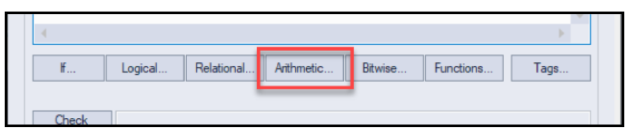




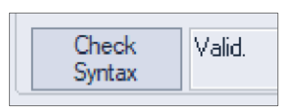
In the Expression dialog, select **Tags...** button and choose the timer Alias tag.

The tag will appear in the **Expression** field

Select **Arithmetic...** button, then appropriate operator, and then type in the constant value for the operation to convert the milliseconds to actual time.



Select Check Syntax button the result should read **Valid**.

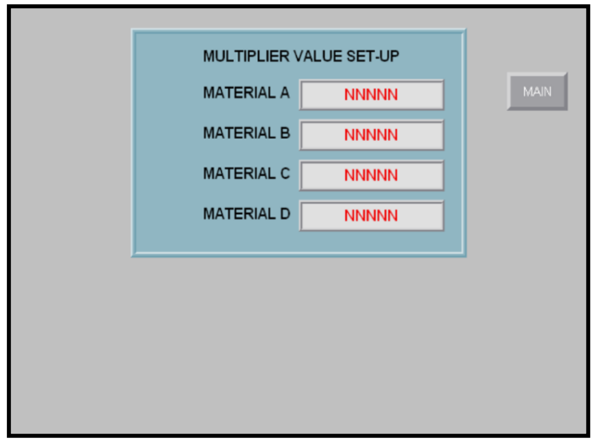


If not check the tag information operator, or value.

Select the **OK** buttons in the dialogs to accept the connections.

continue

OPERATORS PANEL 2



- Delete the unused objects.
- Edit to change the static text captions.
- Edit to change button objects captions.
- Edit the **Numeric Input** objects, in the **Numeric** tab change the following:

  Numeric pop-up: Keypad

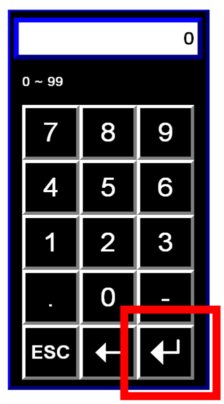
  Maximum value: 10

  Number of digits: 5

  Decimal places: 2
- Edit connection tags to (not text) the numeric input objects.
- Edit the go to navigation for operator's panel 1.
- Format screen appearance to match screen shown.

When selecting Numeric Input during runtime a keypad will appear.

Use cursor to select numbers.



ENTER key

LAB: TEMPERATURE CONVERSION with INDICATORS

This HMI lab will create the .MER Runtime file.

Modify the HMI template TEMPLATE TEMPERATURE PANEL for the TEMPERATURE CONVERSION with INDICATORS lab.

Use Application Manager to make a copy with a new name TEMPERATURE PANEL_*last name*.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures.

Configure communication with Studio 5000 off line-file/processor with a shortcut named **HMI_DATA**.

UNUSED PANELS

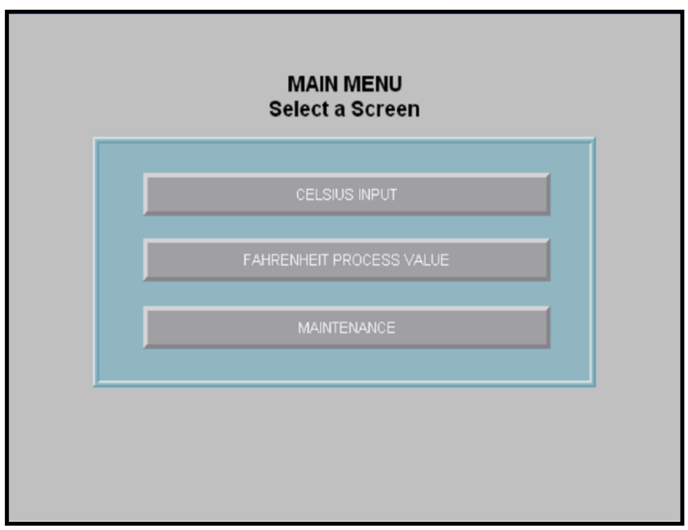
Delete unused panels in the **Explorer** panel – **Displays**.

PLC TAG CONNECTIONS

Assign Connection to appropriate objects on each panel.

OPERATORS PANEL 1

- Delete unused objects (buttons).
- Change the button captions.
- Edit the go to navigation for the required display screens.
- Format screen appearance to match screen shown.



OPERATORS PANEL 2

- Delete unused objects and arrange as shown.
- Edit to change the static text captions.
- Edit to change button objects captions.
- Change error Multistate Indicators when in State 1.

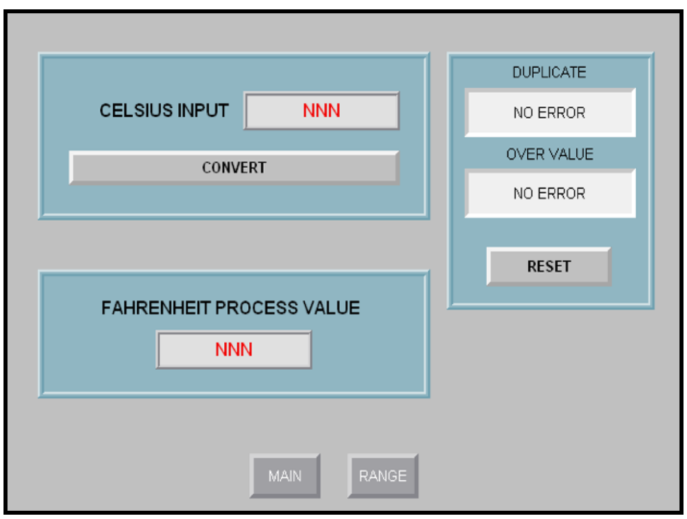
  Set Background to red.

  Change Caption to ERROR with caption color of white and bold.
- Change Celsius Input Numeric Input to three (3) digits, 0 decimal places..

  Set Numeric Input range to 0 to 200.

  Note: Celsius numeric input will not be visible when TWS is enabled.

  See page 11: ANIMATION on Operators Panel 2 for Celsius Input
- Numeric display to three (3) digits, 0 decimal places.
- Edit the go to navigation for the required display screens.
- Format screen appearance to match screen shown.



continue

OPERATORS PANEL 3

- Delete unused objects and arrange as show.
- Edit to change the static text captions.
- Select the Gauge and open the Properties dialog.

  Change Gauge range to 0 to 212.

  Change Major Ticks to 7.

See below To Select a Multistate Indicator.

- Change all Multistate Indicators, State 0 to the gray background color of the panel.
- Change Multistate Indicators, State1 to:

High High	- Bright Red
High	- Dark Red
Med	- Dull Orange
Low	- Cyan
Low Low	- Dark Blue

See illustration for colors on screen below.

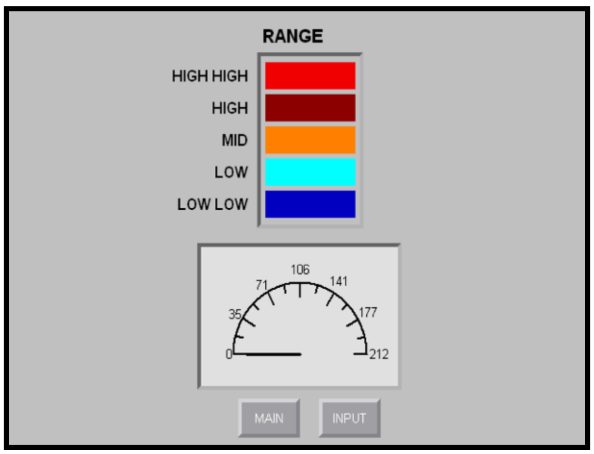
To Select a Multistate Indicator

Place the cursor over indicator object and single left click.
This will select the outside panel.

Without moving the cursor, single left click again.
This will select the indicator.

Alternative: Select from Toolbox the tab for the Object Explorer.

Double left click on object's name in the Object Explorer tree to open the object's properties dialog.

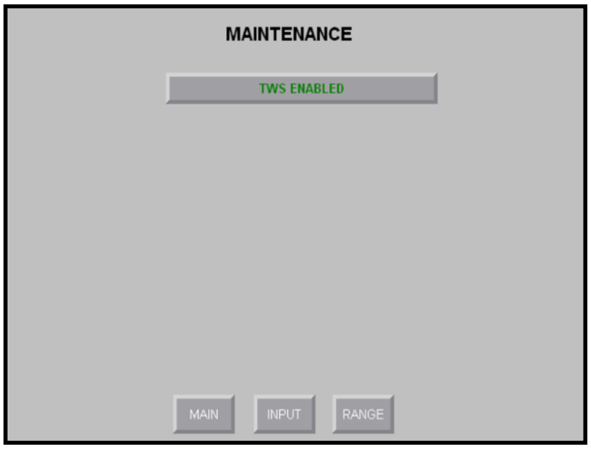


- Edit the go to navigation for the required display screens.
- Format screen appearance to match screen shown.

continue

OPERATORS PANEL 4

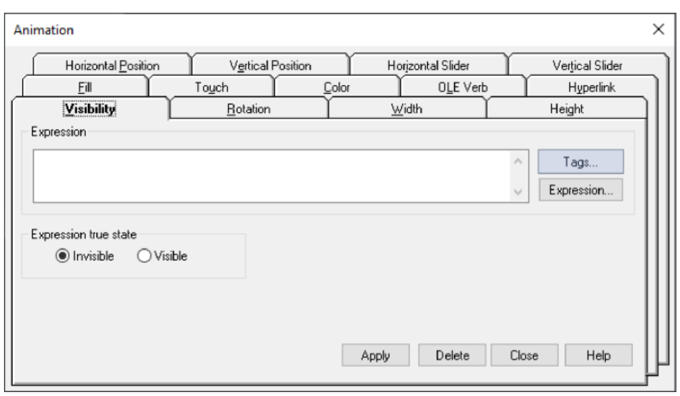
- Delete unused objects and arrange as shown.
- Edit to change the static text captions.
- Change Maintained Button State 0 caption to 'TWS ENABLED' and to Dark Green and Blinking.
- Change Maintained Button State 1 caption to 'TWS DISABLED' and to Bright Red.



ANIMATION on OPERATORS PANEL 2 for CELSIUS NUMERIC

- Select the Numeric Input object on Operator Panel 2.
- From the Menu bar select, Animation and then Visibility.

Alternative: Right click on the Numeric Input and select Animation from the context menu.



Continue

- Select the **Invisible** button.
- Select the **Tags...**, button and from Program:MainProgram to select theTg for the Thumbwheel Switch Enable\Disable.
- At the end of the tag string, type **== 0**.
- Select the **Apply** button.
- Select the **Close** button.

DESIGN TESTING SHORTCUT

Configure communication with Studio 5000 processor for Design testing.

Test the application.

When applications tests correctly, configure communication with Studio 5000 processor for **Runtime**.

RUNTIME SHORTCUT

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 8 SHORTCUT ASSIGNMENT FOR RUNTIME.

RUNTIME APPLICATION FILE

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 11 CREATE RUNTIME APPLICATION.

ME STATION RUNTIME APPLICATION

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.

LAB: ANALOG INPUT

This HMI lab will create the .MER Runtime file.

Modify the HMI template TEMPLATE ANALOG INPUT for the ANALOG INPUT lab.

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures.

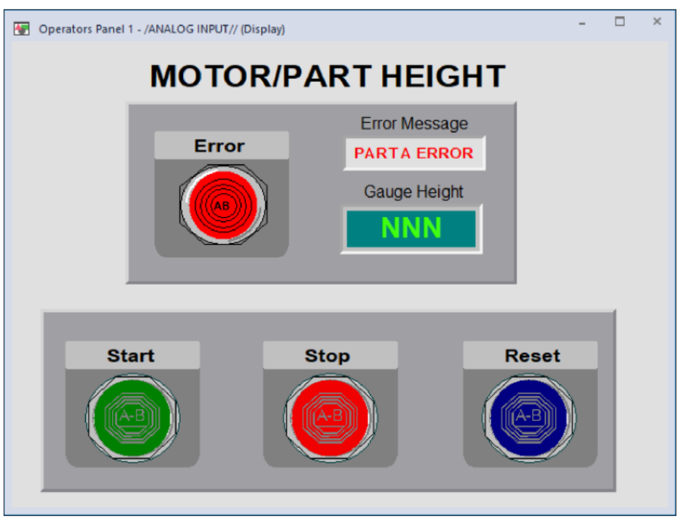
Use Application Manager to make a copy with a new name ANALOG INPUT *your initials*.

Configure communication with Studio 5000 off line-file/processor with a shortcut named **HMI_DATA**.

Format the screen layout as shown.

IMPORTANT FOR GRADE

When formatting, make sure that added Indicator and Pushbutton objects and Overlay objects are centered in relationship with the existing objects associated with the added objects as shown.



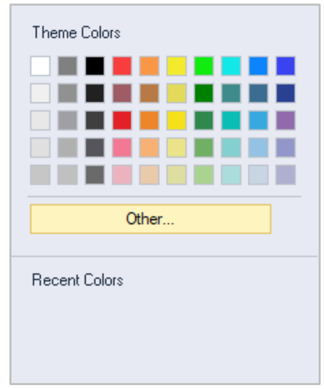
The **START**, **STOP**, and **RESET** pushbuttons and **ERROR** indicator do not exist. You must add those objects to the HMI

INFORMATION ONLY – OTHER COLORS

The Theme Colors palette for an object is a standard Windows dialog.

Not all colors not shown in the palette needed are shown in the color grid.

To select additional colors, select **Other…**.



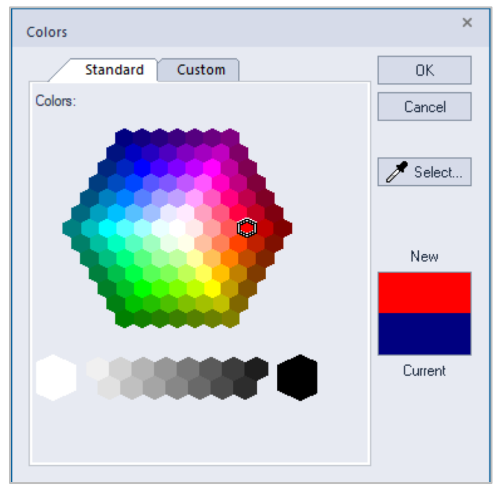
The Colors dialog appears.

Select the **Standard** tab at the top.

Select the required color hexagon on the palette.

Select the **OK** button.

The color is assigned to the object's attribute.

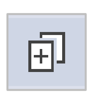


INFORMATION ONLY – DUPLICATE AN OBJECT

To duplicate an object right click on the object and select **Duplicate** from the context menu.

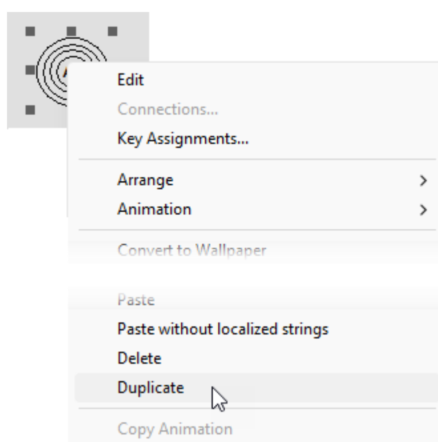
- OR -

Left click on the object and from the toolbar select the **Duplicate** icon



- OR -

Left click on the object and on the keyboard press **Ctrl + D**



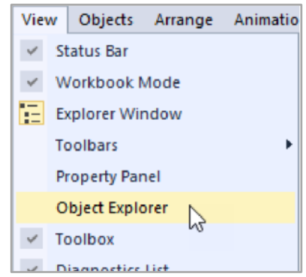
continue

INFORMATION ONLY - OBLECT EXPLORER

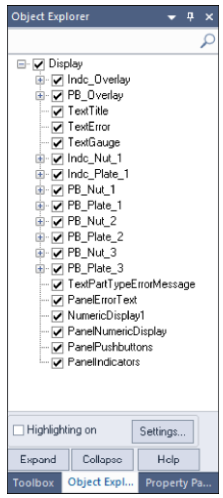
Use the Object Explorer to assist in selecting a graphic object on the display.

A Display must be opened first.

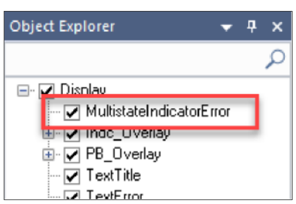
Select from the menu **View** and then **Object Explorer**.



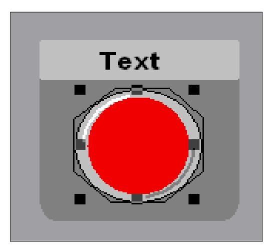
The Object Explorer window will appear.



As objects are added, they will be included in the listing.



An object selected from the list will then have the “handles” around it in the display



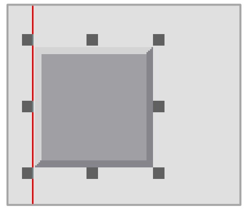
INFORMATION ONLY – KEYBOARD CURSOR KEYS TO MOVE OBJECT

To move a selected object with the keyboard cursor arrow keys the mouse cursor must be over the object.

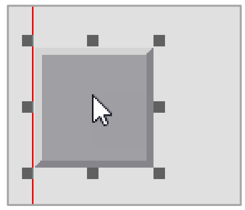
continue

Object is a generic example.

Select the object by with a single left click.

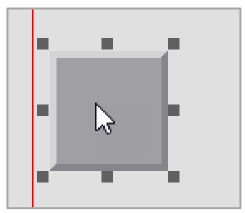


Move the cursor over the object, the mouse key does not need to be held down.



Using keyboard cursor arrow keys move the object one pixel in a direction.

If the object did not move, left click to select.



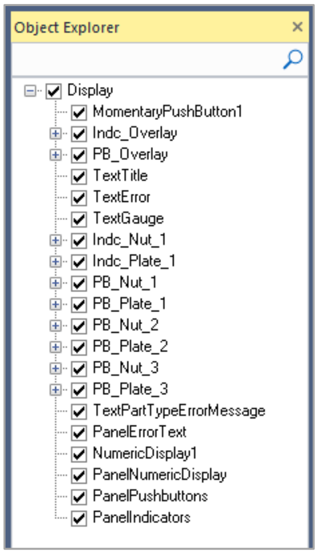
INFORMATION ONLY – EDIT NAMEPLATE TEXT and ALIGNING OBJECTS

To edit the text on the **Indicator** and **Pushbutton** nameplates.

This example is for the **START Text** pushbutton nameplate text, ***n*** is the object's number

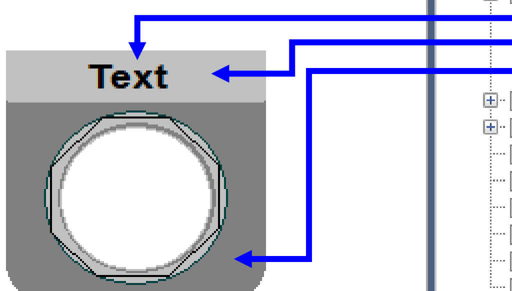
continue

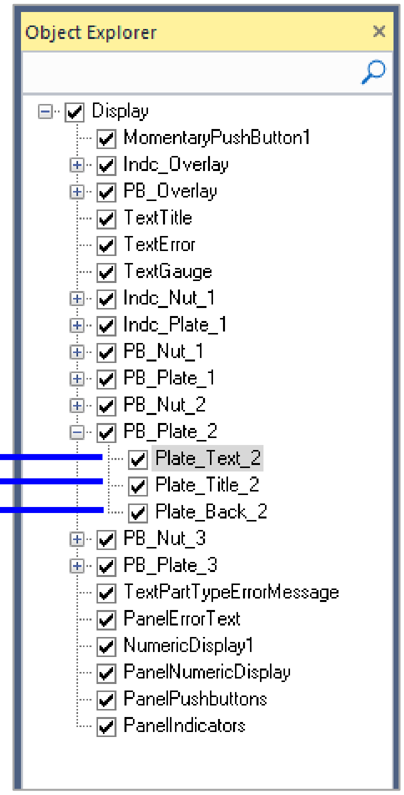
In the Object Explorer select and expand the **PB_Plate_*n*** object.



The expanded object shows the:

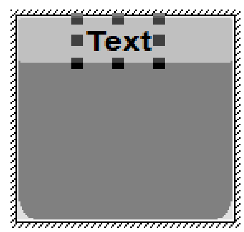
- Nameplate Text
- Nameplate Title Background
- Nameplate Backing



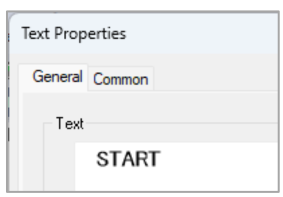


Note: *The Indicator and other Pushbutton Plates use the similar naming conventions.*

In the expanded Object Explorer select the text object, in this example **Plate_Text_2**.



In Properties dialog change the **Text** to the required name.

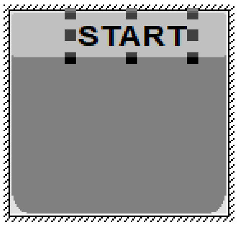


continue

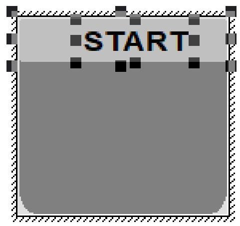
To select multiple objects, hold down the **Ctrl** key while left clicking on the additional object.

With the Text still selected, hold down the **Ctrl** key and left click on the **Plate_Title_2**.

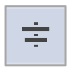
Text Selected



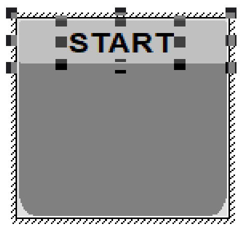
Text and **Plate Title** Selected



Select the **Align Center** button in the toolbar.

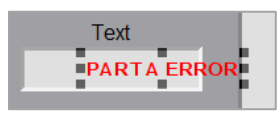


The text is centered on the nameplate.

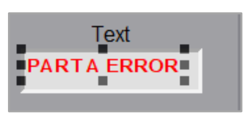


INFORMATION ONLY – EDIT ERROR TEXT and ALIGNING OBJECTS

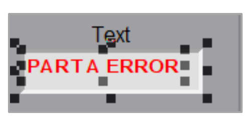
Select the **Text** object.



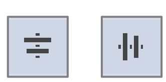
Drag the **Text** object into the Error Message field.



Press and hold the **Ctrl** key and select the **Error Message** field.



Select the **Align Center** and **Align Middle** in the toolbar.



The text is centered in the message field.

INFORMATION ONLY – CENTERING OBJECT

To center the **Indicator**, **Pushbutton**, and Overlays to the plate’s locking nut.

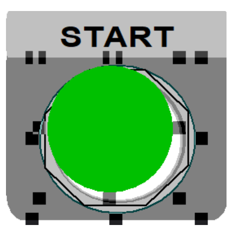
This example is for the **START Pushbutton** and **Overlay**, ***n*** is the object’s number

PUSHBUTTON

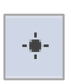
Drag the **Pushbutton** (or **Indicator**) object over the lockdown nut.

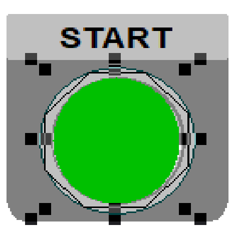
You may need to **Zoom In** to be able to select the lockdown nut.

With the **Pushbutton** still selected, hold down the **Ctrl** key and left click on the **Lockdown Nut**.



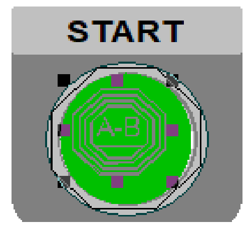
Select the **Align Centers Points** button in the toolbar.



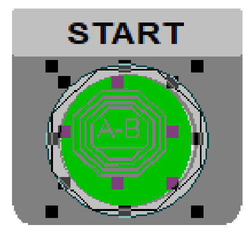


OVERLAY

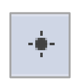
Drag the **Overlay** over the pushbutton.

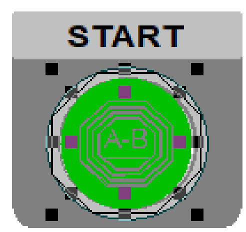


With the **Overlay** still selected, hold down the **Ctrl** key and left click on the **Push-button**.



Select the **Align Center Points** button in the toolbar.





continue

PLC TAG CONNECTIONS

- Assign Connection to appropriate objects on each panel.

When assigning tags, ignore the Subroutine and Tags for Emulation.

EDIT STATIC TEXT

- Edit the static text for the **Title**, **Error Message** field, and **Numeric Display** field.
- Align text as shown on the screen illustration, page 13.

PUSHBUTTON and INDICATOR OBJECTS

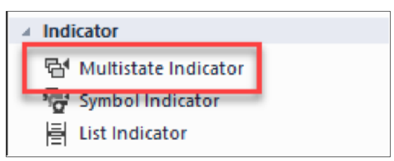
The **START**, **STOP**, and **RESET** pushbuttons and **ERROR** indicator do not exist. You must add those objects to the HMI.

Select the **Toolbox** tab at the bottom of the docked panels on the right side of the screen.

All added objects are:
Shape: Circle
Back style: Solid
Border style: Line

Error Indicator

Number of states: 2
State 0: Dark Red
State 1: Bright Red
Width: 66
Height: 66
Name: MultistateIndicatorError

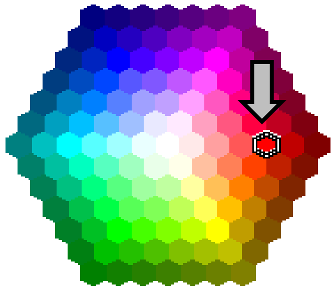


- Make a duplicate of the embossed circle logo, **Indc_Overlay**, and drag over the indicator and align.

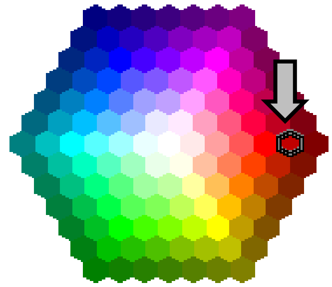
By Duplicating, the new Indc_Overlay *will be on the top layer over the indicator.*

- Assign the Connection.
- State Back Colors

State 0: Bright Red

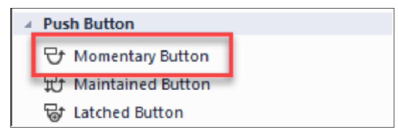


State 1: Dark Red



Start Pushbutton

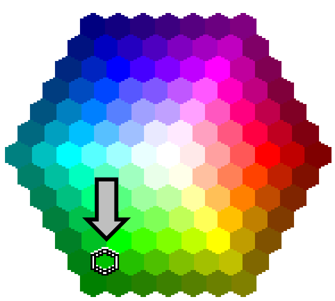
Button action: Normally Open
State 0: Green
State 1: Dark Green
Width: 74
Height: 74
Name: MomentaryPushButtonStart



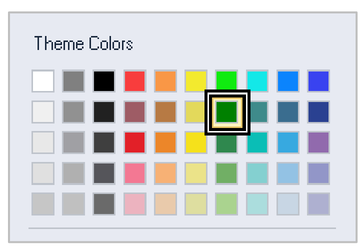
State 1: Dark Red

- Make a duplicate of the embossed octagon logo, **PB_Overlay**, and drag over pushbutton and align.
- Assign the Connection.
- State Back Colors

State 0: Green



State 1: Dark Green

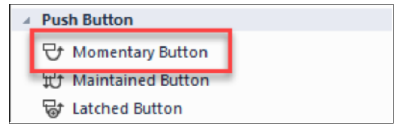


---

continue

Stop Pushbutton

Button action: Normally Closed
State 0: Red
State 1: Dark Red
Width: 74
Height: 74
Name: MomentaryPushButtonStop



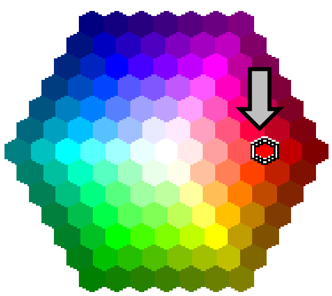
- Make a duplicate of the embossed octagon logo, **PB_Overlay**, and drag over pushbutton and align.

- Assign the Connection.

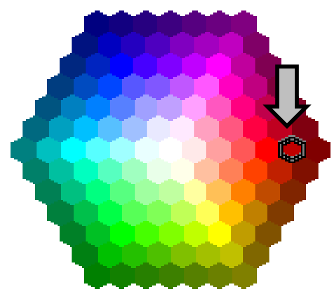
continue to color palette illustrations

- State Back Colors

State 0: Bright Red

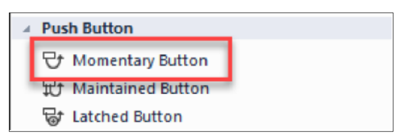


State 1: Dark Red



Reset Pushbutton

Button action: Normally Open
State 0: Blue
State1 : Dark Blue
Width: 74
Height: 74
Name: MomentaryPushButtonReset



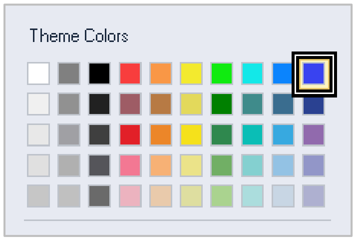
- Make a duplicate of the embossed octagon logo, PB_Overlay, and drag over pushbutton.

- Assign the Connection.

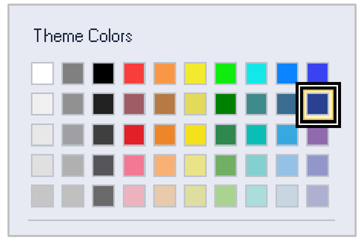
- Make a duplicate of the embossed octagon logo, **PB_Overlay**, and drag over pushbutton and align.

- State Back Colors

State 0: Blue



State 0: Dark Blue



- Delete the original **Indc_Overlay** and **PB_Overlay**.

ERROR MESSAGE TEXT

- Make a duplicate of the **TextPartTypeErrorMessage**.

  Text (Part A): Text: PART A ERROR
  (Re)Name: TextPart**A**TypeErrorMessage

  Text (Part B) Text: PART B ERROR
  (Re)Name: TextPart**B**TypeErrorMessage

- Set the Expression in the Animation Property to **Visible** for the text for part type error message.

- Align the Text with the background field.

  Normally, the text is invisible. When the assigned tag is set to 1, the expression sets the text to visible.

NOTE: When creating the animation for the text, don't forget to use the double equals, ==, when evaluating the Tag for a value of one.

---

continue

NUMERIC DISPAY

- Assign the Connection.

DESIGN TESTING SHORTCUT

Configure communication with Studio 5000 processor for Design testing.

Test the application.

When applications tests correctly, configure communication with Studio 5000 processor for **Runtime**.

RUNTIME SHORTCUT

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 8 SHORTCUT ASSIGNMENT FOR RUNTIME.

RUNTIME APPLICATION FILE

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 11 CREATE RUNTIME APPLICATION.

ME STATION RUNTIME APPLICATION

See desktop PDF document FactoryTalk View Studio Machine Edition Procedures on page 15 Appendix B: STATION RUNTIME APPLICATION.